6.9.1 The coordinates of the image of a point in a plane

Let (a_1, a_2, a_3) be the point $\vec{u}$ whose image in the plane is required. Then $\vec{u} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$.

Let $ax + by + cz = d$ be the equation of the given plane.Writing the equation in the vector form we get $\vec{r} \cdot \vec{n} = p$ where $\vec{n} = a\hat{i} + b\hat{j} + c\hat{k}$ .Then the position vector of the image is

$$\vec{v} = \vec{u} + \frac{2\left[p - (\vec{u} \cdot \vec{n})\right]}{|\vec{n}|^2}\vec{n}\,.$$

If $\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$, then $v_1 = a_1 + 2a\alpha$, $v_2 = a_2 + 2a\alpha$, $v_3 = a_3 + 2a\alpha$

$$\text{where } \alpha = \frac{2\left[p - (aa_1 + ba_2 + ca_3)\right]}{a^2 + b^2 + c^2}.$$

Example 6.55

Find the image of the point whose position vector is $\hat{i} + 2\hat{j} + 3\hat{k}$ in the plane $\vec{r} \cdot \left(\hat{i} + 2\hat{j} + 4\hat{k}\right) = 38$.

Solution

Here, $\vec{u} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{n} = \hat{i} + 2\hat{j} + 4\hat{k}$, $p = 38$. Then the position vector of the image $\vec{v}$ of

$\vec{u} = \hat{i} + 2\hat{j} + 3\hat{k}$ is given by $\vec{v} = \vec{u} + \dfrac{2\left[p - (\vec{u} \cdot \vec{n})\right]}{|\vec{n}|^2}\vec{n}$.

$$\vec{v} = \left(\hat{i} + 2\hat{j} + 3\hat{k}\right) + \frac{2\left[38 - \left(\left(\hat{i} + 2\hat{j} + 3\hat{k}\right) \cdot \left(\hat{i} + 2\hat{j} + 4\hat{k}\right)\right)\right]}{\left(\hat{i} + 2\hat{j} + 4\hat{k}\right) \cdot \left(\hat{i} + 2\hat{j} + 4\hat{k}\right)}\left(\hat{i} + 2\hat{j} + 4\hat{k}\right).$$

That is, $\vec{v} = \left(\hat{i} + 2\hat{j} + 3\hat{k}\right) + 2\left(\dfrac{38 - 17}{21}\right)\left(\hat{i} + 2\hat{j} + 4\hat{k}\right) = 3\hat{i} + 6\hat{j} + 11\hat{k}$.

Therefore, the image of the point with position vector $\hat{i} + 2\hat{j} + 3\hat{k}$ is $3\hat{i} + 6\hat{j} + 11\hat{k}$.

Note

The foot of the perpendicular from the point with position vector $\hat{i} + 2\hat{j} + 3\hat{k}$ in the given plane is

$$\frac{(\hat{i} + 2\hat{j} + 3\hat{k}) + (3\hat{i} + 6\hat{j} + 11\hat{k})}{2} = 2\hat{i} + 4\hat{j} + 7\hat{k}\,.$$

6.10 Meeting Point of a Line and a Plane

Theorem 6.23

The position vector of the point of intersection of the straight line $\vec{r} = \vec{a} + t\vec{b}$ and the plane $\vec{r} \cdot \vec{n} = p$ is $\vec{a} + \left(\dfrac{p - (\vec{a} \cdot \vec{n})}{\vec{b} \cdot \vec{n}}\right)\vec{b}$, provided $\vec{b} \cdot \vec{n} \neq 0$.

Proof

Let $\vec{r} = \vec{a} + t\vec{b}$ be the equation of the given line which is not parallel to the given plane whose equation is $\vec{r} \cdot \vec{n} = p$. So, $\vec{b} \cdot \vec{n} \neq 0$.

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